

9. Apache Groovy

According to its website, “Apache Groovy is a powerful, optionally typed and dynamic language, with static-typing and static compilation powerful features, for the Java platform aimed at improving developer productivity, thanks to a concise, familiar and easy-to-learn syntax. It integrates smoothly with any Java program, and immediately delivers to your application powerful features, including scripting capabilities, domain-specific language authoring, runtime and compile-time meta-programming and functional programming.”

Apache Groovy is a dynamic language and has features similar to other programming languages like Python, Ruby, Perl and Smalltalk. Groovy also supports modularity, type checking, static compilation, Project Coin syntax enhancements, multi-catch blocks and ongoing performance enhancements using JDK7's invoke dynamic instruction. Groovy provides native support for various mark-up languages such as XML and HTML, accomplished via an inline DOM syntax. This feature enables the definition and manipulation of many types of heterogeneous data assets with a uniform and concise syntax and programming methodology.

The latest version of Apache Groovy is 2.4.7

The syntax for the ‘Hello World’ program is:

```
class Hello {
    String name;
    void say Hello() {
        System.out.println("Hello "+get Name()+"!");
    }
    void set Name(String name) {
        pathnames = name;
    }
    String get Name() {
        return name;
    }
    static void main(String[] args) {
        Hello hello = new Hello();
        Hello.hello("world");
        Hello.hellos();
    }
}
```

The advantages of Apache Groovy

- Apache Groovy is easy to learn and code, especially for Java developers, as it contains closures, builders, runtime and compile time meta programming, functional programming, type inference and static compilation.
- Groovy has a rich and vibrant ecosystem in terms of Web development, reactive applications, concurrency, parallelism library, test frameworks, build tools, code analysis and GUI building.
- Groovy supports smooth Java integration, and also interoperates with Java and other third party libraries.

The disadvantages

- When used with lots of dynamic features like duck-typing, dynamic code, meta-programming, etc, Groovy demands more processing at runtime and this results in a slow performance.
- Groovy lacks documentation as well as online resources, so there's not much help for newbies. Also, it has not been implemented in high-end and critical applications. 

References

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